

RESEARCH + MENTORSHIP @ SIRE Lab

RESEARCH. **EXPOSURE.** EXPERIENCE. **IMPACT.**

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PI SIRE Lab

MORGAN STATE UNIVERSITY

Civil Engineering Department



SUSTAINABLE INFRASTRUCTURE DEVELOPMENT,
SMART INNOVATION AND RESILIENT
ENGINEERING RESEARCH LAB

*Advancing Sustainable, Smart, and Resilient
Infrastructure for a better future.*



OUR MISSION

To advance sustainable, smart, and resilient infrastructure through interdisciplinary research, education, innovation, and partnerships that improve infrastructure performance, strengthen community resilience, and support sustainable development.



DIGITAL TWINS &
INFRASTRUCTURE MODELS



SMART
TRANSPORTATION



SUSTAINABLE &
RESILIENT INFRASTRUCTURE

THE PREMISE

THE STORY

Student Success follows engineering scholars as research becomes a public practice—presented, challenged, refined and carried into the field.

19

RESEARCH STUDENTS



THE JOURNEY

FOUR ENVIRONMENTS • ONE CONTINUOUS STORY

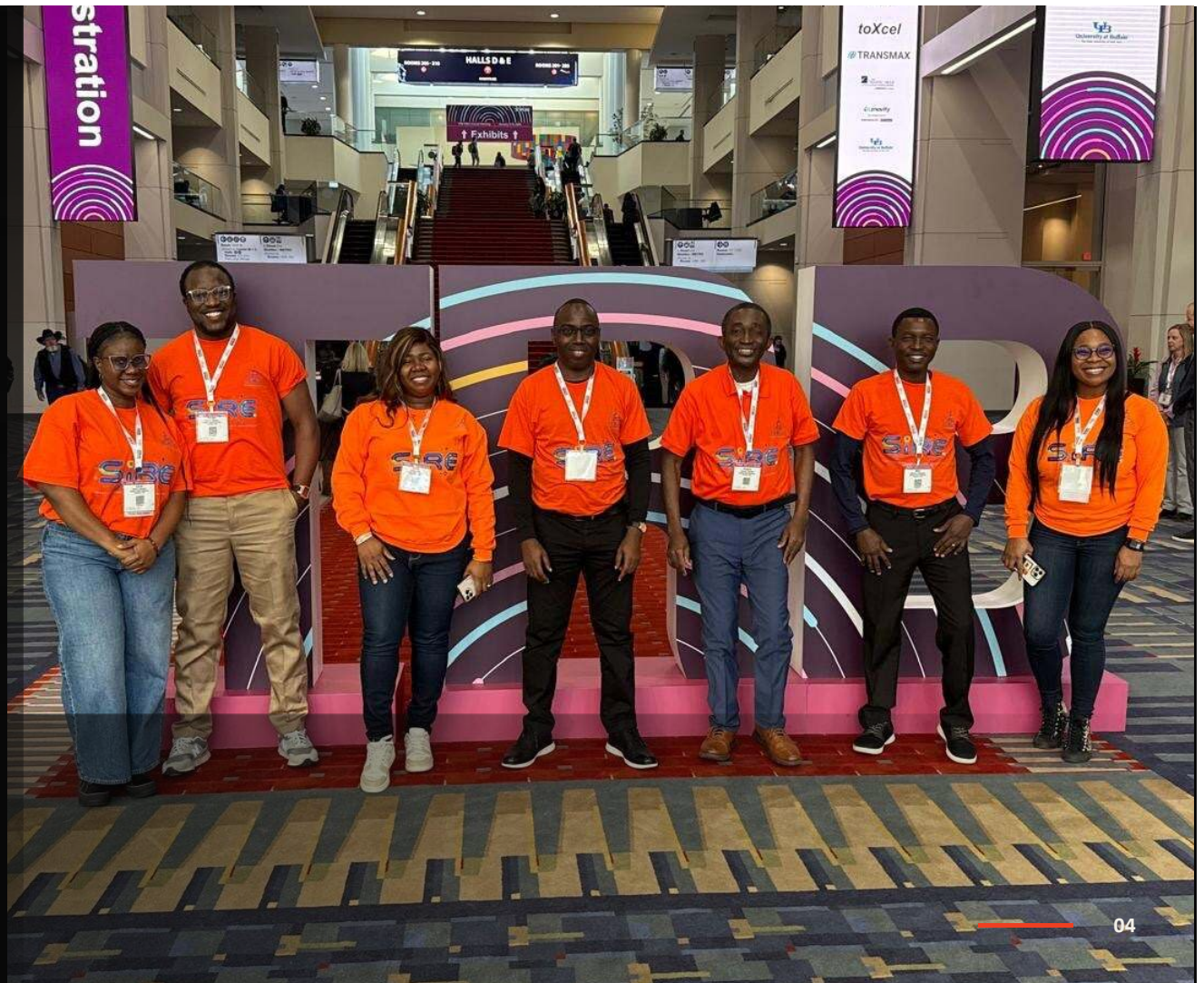


ACTIVITY 01

WASHINGTON, D.C.

TRB 2026

19 RESEARCH STUDENTS





THE TEAM

19 SCHOLARS • ONE RESEARCH COMMUNITY

THE NETWORK

CONNECTIONS BUILT ACROSS DISCIPLINES, INSTITUTIONS AND GENERATIONS.



PROFESSIONAL CONNECTIONS



RESEARCH COMMUNICATION

THE STAGE

Students move from preparation to public presentation—building confidence before a live professional audience.



THE POSTER FLOOR

IDEAS BECOME CONVERSATIONS



A photograph of Michael Okegbola, a man with glasses wearing a dark suit and a white shirt, standing behind a wooden podium. The podium has a blue sign with the letters 'TRB' on it. He is in a room with a plain wall and a door to his right. A red fire alarm pull station is visible on the wall. The image is dimly lit, with the speaker and podium being the main focus.

PAPER PRESENTATION • TRB

MICHAEL OKEGBOLA

A research idea enters the room, meets an audience and becomes part of the professional conversation.

ACTIVITY 02

NEW ORLEANS • DECEMBER 2025

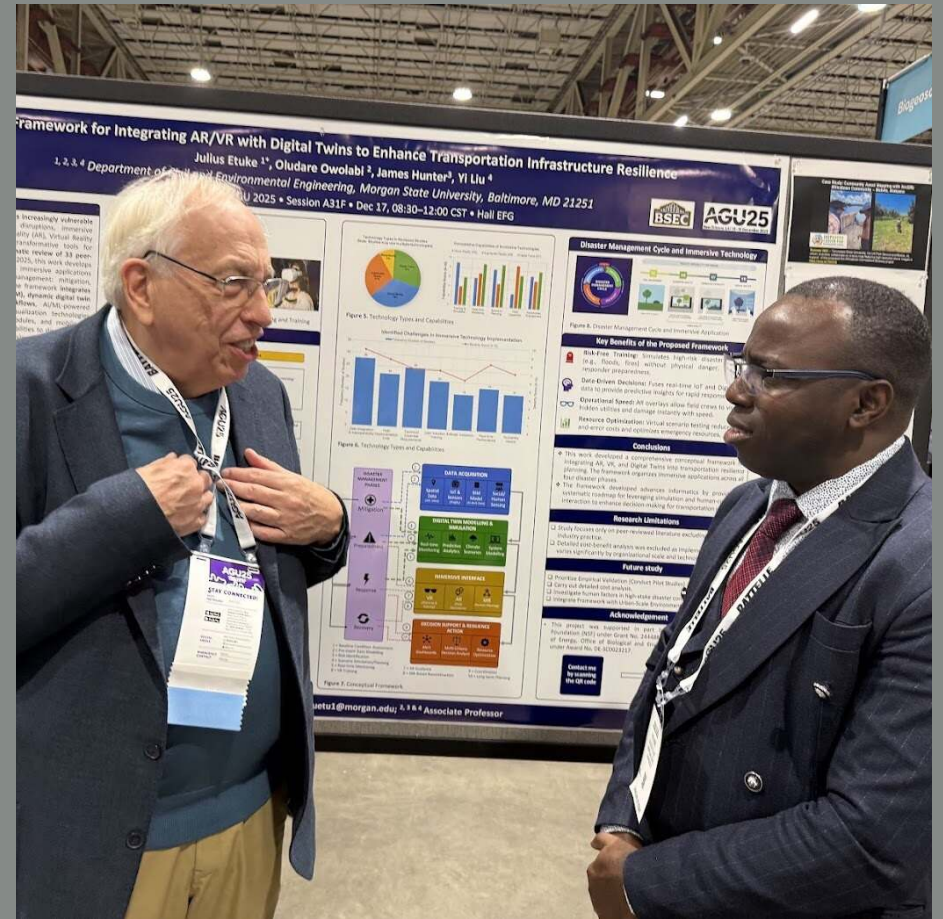
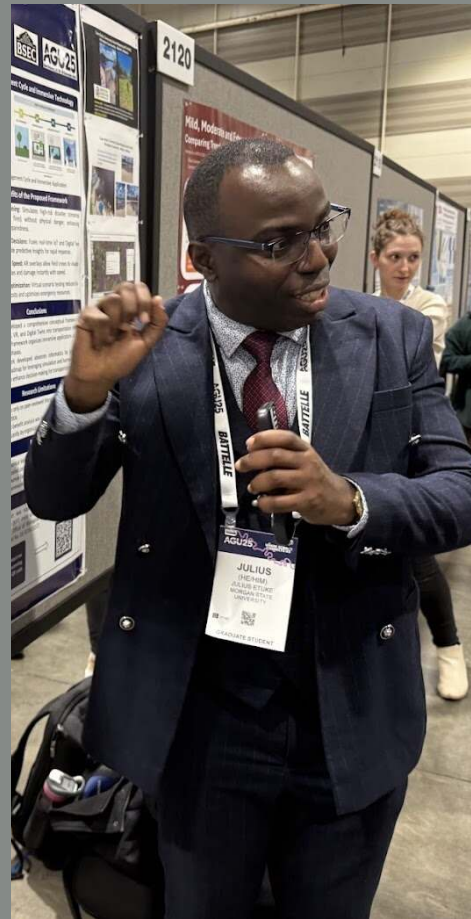
AGU

RESEARCH ON A NATIONAL STAGE

POSTER PRESENTATION • AGU

JULIUS ETUKE

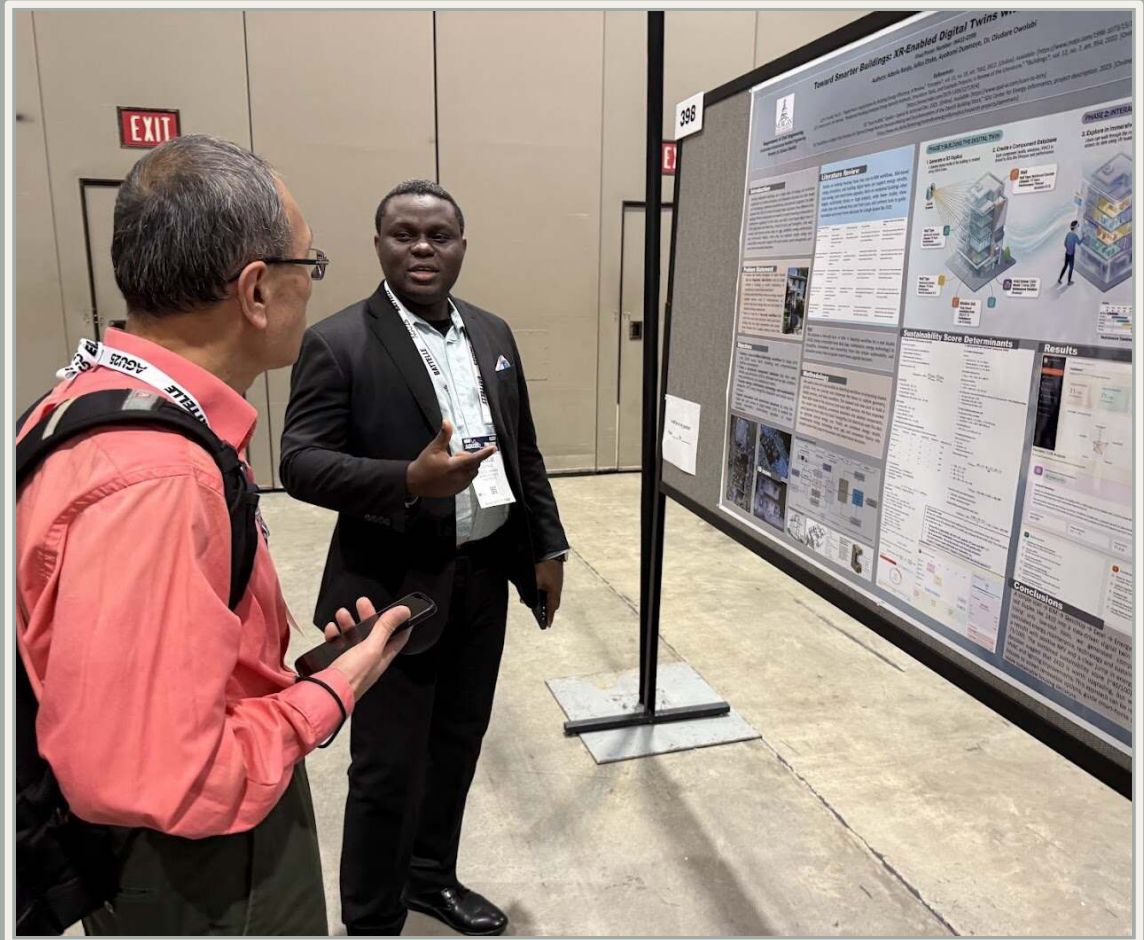
Explaining the work. Testing the argument. Building a new professional network around the research.



DIGITAL TWIN • AGU 2025

ADEOLU BANJO

Technical depth becomes visible
when a student can make complex
research clear to a new audience.

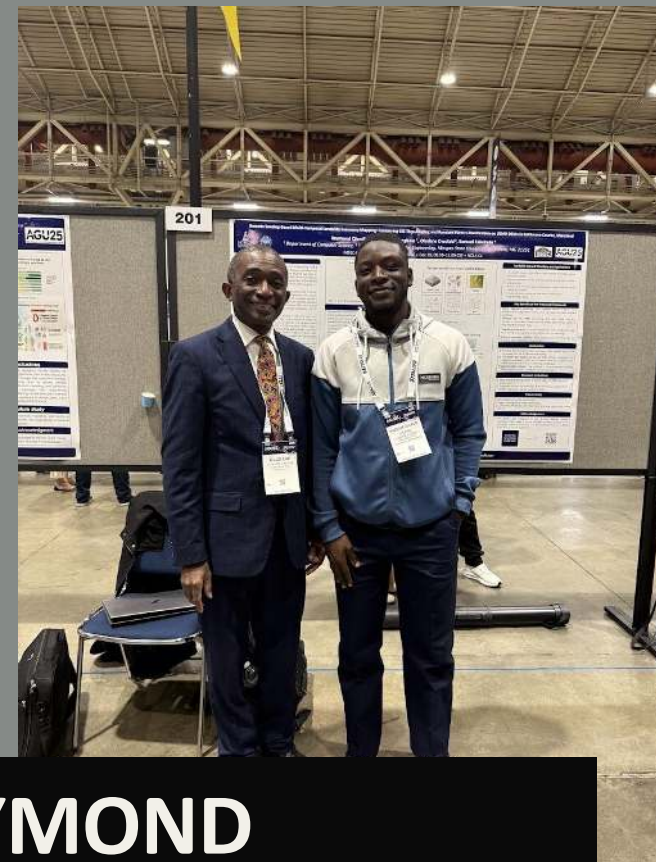
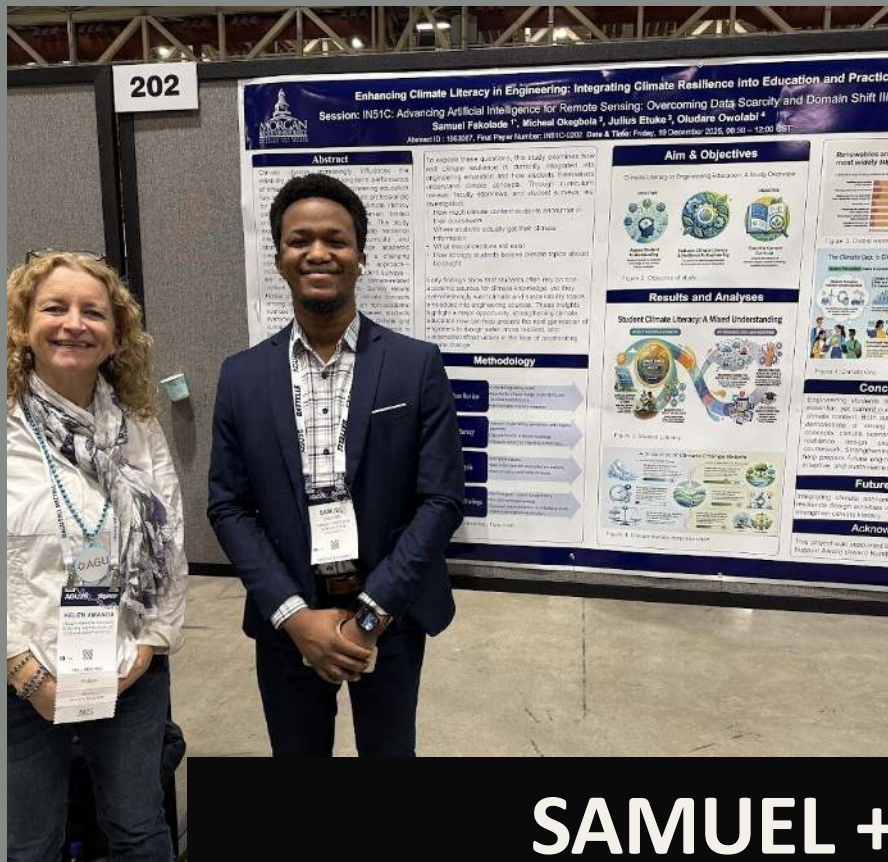




POSTER PRESENTATION

AYOBAMI DUNMOYE

RESEARCH IN DIALOGUE
WITH A NATIONAL AUDIENCE.

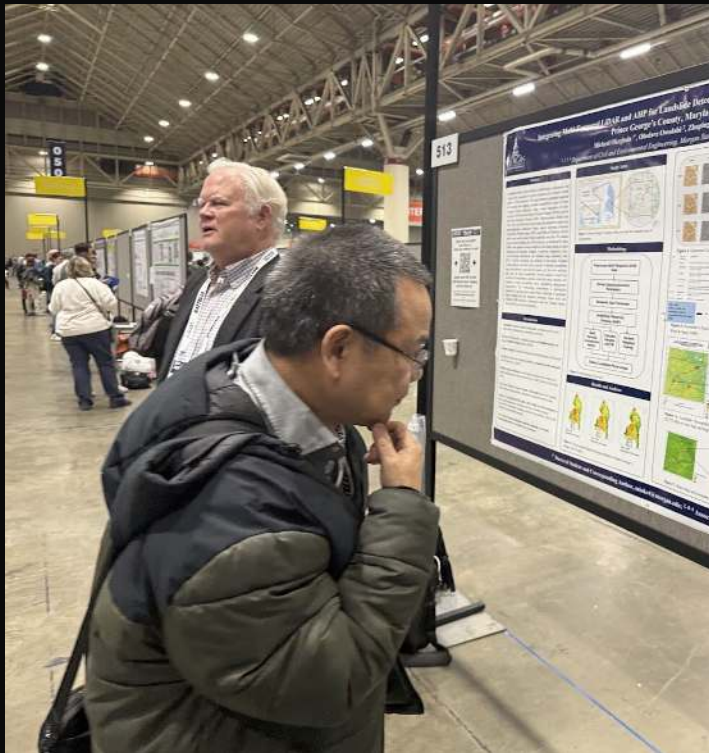


SAMUEL + RAYMOND

TWO RESEARCH STORIES • ONE GLOBAL STAGE

Collaborations at AGU

FROM PAPER PRESENTATION TO POSTER EXCHANGE



RESEARCH • EXCHANGE • VISIBILITY



THE AGU TEAM

RESEARCH ON A NATIONAL STAGE



ACTIVITY 03

OCTOBER 2025



COLLEGE OF ENGINEERING
Aerospace, Chemical, Computer, Electrical,
Materials & Mechanical

TUSKEGEE UNIVERSITY

TUSKEGEE

ACADEMIC EXCHANGE

THE VISIT

TUSKEGEE UNIVERSITY • OCTOBER 2025



SITE-BASED ENGAGEMENT

LEARNING ON SITE

Seeing facilities, tools and
decisions up close turns abstract
knowledge into lived professional
experience.



A group of approximately 15 young men and one older man are sitting on a low stone wall in front of a large bronze statue of Booker T. Washington. The statue is seated and holds a book. The base of the statue is inscribed with "BOOKER T. WASHINGTON 1856-1915" and a quote: "HE OPENED THE WAY OF KNOWLEDGE FOR THE PEOPLE AND POINTED THE WAY TO BRING THE EDUCATION AND INDUSTRY". The group consists of 14 young men wearing orange t-shirts with a logo, and one man in a white jacket. They are all smiling and looking towards the camera. The background features trees with autumn foliage and a clear sky.

HISTORY • IDENTITY • COMMUNITY

LEGACY + COMMUNITY

Shared experience connects today's students to institutional history—and to one another.

ACTIVITY 04

AUBURN, ALABAMA • OCTOBER 2025

NCAT

FIELD-BASED LEARNING

National Center
For
Asphalt Technology



NCAT • AUBURN

INSIDE THE LAB`

Applied learning comes
alive around materials,
testing equipment and the
people who use them
every day.





THE FACILITY



RESEARCH FACILITIES IN PRACTICE



ACTIVITY 05

BEYA CONFERENCE • 2026



BEYA

Civil Engineering for Kids



BEYA

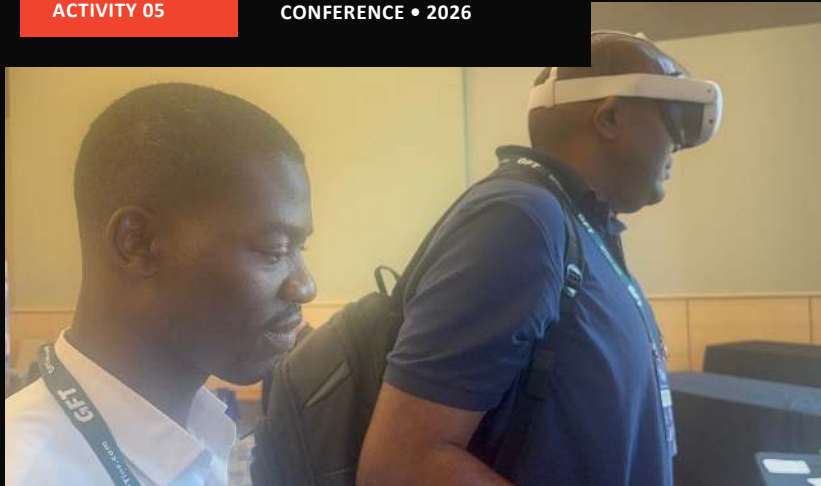
Civil Engineering for Kids

ACTIVITY 05

CONFERENCE • 2026

GIS TOWSON

GEOSPATIAL RESEARCH IN MOTION



GIS TOWSON

GEOSPATIAL RESEARCH IN MOTION

ACTIVITY 06

SYMPOSIUM • 2026



NSEA (Morgan)

ENGINEERING MEETS INTELLIGENCE

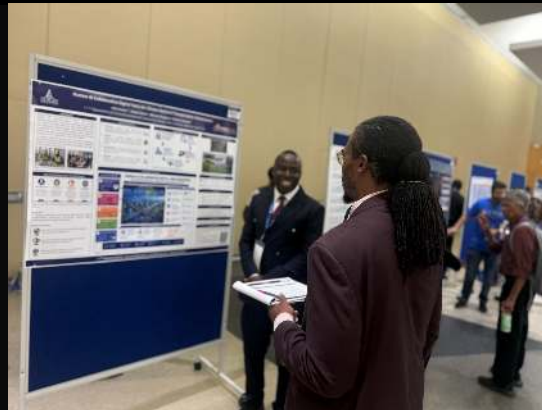
ACTIVITY 06

SYMPOSIUM • 2026



NSEA (Morgan)

ENGINEERING MEETS INTELLIGENCE



ENGINEERING INNOVATION IN ACTION

SIRE RESEARCH LABORATORY

Sustainable Infrastructure Development, Smart Innovation, and Resilient Engineering

Morgan State University | Engineering Futures Workshop | July 2026



ACTIVITY 07

INDUSTRY • 2026

AFRO EXPO

INNOVATION + COMMUNITY

ACTIVITY 07 INDUSTRY • 2026



AFRO EXPO

INNOVATION + COMMUNITY

ACTIVITY 08

ENGINEERING • 2026

YOUTH WEEK

DISCOVERY STARTS EARLY

ACTIVITY 08

ENGINEERING • 2026



CATRES (NASA)

INFRASTRUCTURE IN THE FIELD

ACTIVITY 08

AVIATION ENGINEERING • 2026



CATRES (NASA)

DISCOVERY STARTS EARLY



ACTIVITY 09

SITE VISIT • 2026



DFW AIRPORT

INFRASTRUCTURE IN THE FIELD





ACTIVITY 11

ASEE • 2026

DIGITAL TWIN

IMMERSIVE ENGINEERING IN PRACTICE

ACTIVITY 12

ASEE • 2026

and the story continues....

RESEARCH. MENTORSHIP. EXPOSURE. IMPACT.

INTRODUCTION/BACKGROUND

Sustainability has become a critical component of engineering education, and design programs increasingly emphasize that students should include environmental and societal impacts in their design processes. However, the charge to consider sustainability is often without strong evaluative tools, student often resort to features, like bolting on solar panels or changing the material of construction, without conducting a trade-off analysis worth considering, whereas emerging technologies can be used for training engineering students on innovative trial-and-error analysis concepts for various practical systems of their respective fields of engineering.

RESEARCH GOAL AND OBJECTIVES

This research aims to bridge the gap between sustainable design intent and measurable decision-making through the design of a digital twin instructional model that visualizes the consequences of design decisions and the ability to analyze trade-offs and justify design choices

METHODOLOGY



Best DEI paper Award